# INTRODUCTION

Buffet is a dynamic behavior of aircraft flying at a high angle of attack. It occurs in the transonic region where the aircraft has substantial forward speed and the angle of attack becomes large due to certain maneuvers initiated by the pilot. The initial phase of buffet is called buffet onset when noticeable vibrations or oscillations are observed. When the angle of attack is further increased, the aircraft responds to the dynamic buffet pressure with severe structural vibrations which are usually coupled with rigid body motion oscillations or control system-induced oscillations. This latter phenomenon of sustained, large-amplitude aircraft buffeting is called wing rock.

The major cause of aircraft buffet in the transonic region is flow separation on top of the wing surface when the angle of attack reaches a certain amplitude. The separated flow may be induced by a shock or by other disturbances. The dynamic loads exerted on the aircraft may be due to the instability of the shock (in intensity and location), or due to the dynamic pressure components of the turbulent flow especially in the separated region, or both. Furthermore, the separated flow wake may engulf the tail sections, which would cause additional instability in the aircraft responses and deterioration of the flight control qualities.

Earlier work on aircraft buffeting may be traced to the associated subject of stalling flutter (References 1, 2, 3). Both stalling flutter and aircraft buffeting are caused by flow separation at high angle of attack. In stalling flutter, the airfoil motion (for instance, torsional motion) contributes to the cyclic flow pattern which causes a sustained oscillation. In aircraft buffeting, the structural elasticity usually is not a major contributing factor (at least in the initial phase of the transonic maneuver). The buffeting loads are aerodynamic in origin; they exist even if the aircraft, or aircraft model, is perfectly rigid.

The subject of aircraft buffeting started to attract the attention of designers and research workers in the 1950's when aircraft development reached a stage such that transonic flights and maneuvers became a routine matter. Typical works by Huston and associates appearing at that time are given as References 4 through 7. Wind tunnel investigations on airfoil or aircraft buffet in this period include those by Humphreys, Coe, Sutter, and associates (References 8, 9, 10, 11). A corresponding work carried out by Pearcey is given as Reference 12. Pearcey also wrote a very informative article on shock-induced separation and buffet loads (Reference 13).

Starting in 1960, research and development on aircraft buffeting were conducted at an accelerated pace. A large number of papers and reports were published dealing with the various aspects of aircraft buffet, including theoretical treatment (e.g., References 14, 15), flight test evaluation (References 16, 17, 18), and wind tunnel testing (References 19 through 24). Still more recent works dealing with buffet of present-day and future aircraft are listed as References 25 through 33.

The work described in this paper deals with the flight test and data evaluations of the Northrop F-5A aircraft in transonic maneuver. A unique feature of the test program involves the extensive dynamic pressure instrumentation using miniaturized semiconductor-type transducers. Both fluctuating buffet